Power Conversion System (PCS) External Modbus Communication Protocol

1725kW | 2.5 MW | 3.5 MW

Revision History

Version	Date	Description	Revised By	Reviewed By
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1 Overview

This protocol applies to communication between **MPCS series** PCS from Wattox and the energy management system (EMS). Modbus Remote Terminal Unit (RTU) or Modbus TCP/IP is adopted. This protocol enables real-time reading of PCS operational data, related statuses, and control command settings.

2 Physical Interfaces

2.1 Modbus RTU Communication

(1) Serial link transmission

- Interface: RS485
- Baud rate: 9600, 19200, 28400, 115200, 460800, or 921600. Default: 115200 bps.
- Start bit: 1
- Data bit: 8
- Parity bit: None
- Stop bit: 1

(2) Data transmission

- Modbus data is read and written by unit of word. High-order bytes are transmitted before low-order bytes.

(3) Frame check

- The Modbus CRC16 method is used for frame check. Low-order bytes are transmitted before high-order bytes.

(4) Device address

- The range of device address is 1 to 99.

*** The specific communication mode is determined by the external interface of the PCS.**

3 Frame Format

3.1 Modbus RTU

(1) 0x03: Read multiple holding registers

Master:

Slave Address	Param. (0x03)	Register Address	Number of Registers	Check (CRC16)
1 byte	1 byte	2 bytes	2 bytes	2 bytes

Normal slave response:

Slave Address	Param. (0x03)	Number of Bytes	Register Value	Check (CRC16)
1 byte	1 byte	1 byte	(2 x N[1]) bytes	2 bytes

Abnormal slave response:

Slave Address	Error Code (0x83)	Exception Code (0x01, 0x02, 0x03, or 0x04)	Check (CRC16)
1 byte	1 byte	1 byte	2 bytes

(2) 0x06: Write single register

Master:

Slave Address	Param. (0x06)	Register Address	Register Value	Check (CRC16)
1 byte	1 byte	2 bytes	2 bytes	2 bytes

Normal slave response:

Slave Address	Param. (0x06)	Register Address	Register Value	Check (CRC16)
1 byte	1 byte	2 bytes	2 bytes	2 bytes

Abnormal slave response:

Slave Address	Error Code (0x86)	Exception Code (0x01, 0x02, 0x03, or 0x04)	Check (CRC16)
1 byte	1 byte	1 byte	2 bytes

Note:

[1] N indicates the number of registers.

[2] High-order bytes are transmitted before low-order bytes, except for CRC check.

(3) Exception code definition

Exception Code	Modbus Definition
0x01	Invalid function
0x02	Invalid data address
0x03	Invalid value
0x04	Slave device fault

4 Query Parameters (Remote Measurement and Remote Signaling)

4.1 PCS Internal Parameters

Param.	Name	Address	Attribute	Data Type	Value Range	Unit	Description
A01	Primary ARM version	0x2F00	RO	U16	0 to 655.35	0.01	XXX.XX
A02	Temporary version of primary ARM	0x2F01	RO	U16	0 to 655.35	0.01	XXX.XX
A03	Secondary ARM version	0x2F03	RO	U16	0 to 655.35	0.01	XXX.XX
A04	Temporary version of secondary ARM	0x2F04	RO	U16	0 to 655.35	0.01	XXX.XX
A05	FPGA version	0x2F05	RO	U16	0 to 655.35	0.01	XXX.XX
A06	FPGA temporary version	0x2F06	RO	U16	0 to 655.35	0.01	XXX.XX
A07	Parameter list version	0x2F07	RO	U16	0 to 655.35	0.01	
A08	HMI version V	0x090E	RO	U16	0 to 655.35	0.01	*Write when HMI is powered on*
A09	Rated power	0x2F08	RO	U16	0 to 65535	1 kW	-
A10	Rated voltage	0x2F09	RO	U16	0 to 65535	1 V	-
A11	Rated current	0x2F0A	RO	U16	0 to 65535	1 A	-
A12	Maximum charging power	0x0123	RO	S16	–32768 to 0	1 kW	-
A13	Maximum discharging power	0x0124	RO	U16	0 to 32767	1 kW	-
A14	Reactive power upper limit	0x0126	RO	U16	0 to 32767	1 kVar	-
A15	Reactive power lower limit	0x0125	RO	S16	–32768 to 0	1 kVar	-
A16	Maximum charging current	0x0127	RO	S16	–32768 to 0	1 A	-
A17	Maximum discharging current	0x0128	RO	U16	0 to 32767	1 A	-
A18	Upper voltage limit of the off-grid line	0x2F0F	RO	U16	0 to 6553.5	0.1 V	-
A19	Lower voltage limit of the off-grid line	0x2F10	RO	U16	0 to 6553.5	0.1 V	-
A20	Upper voltage limit in constant voltage mode	0x0117	RO	U16	0 to 6553.5	0.1 V	
A21	Lower voltage limit in constant voltage mode	0x0118	RO	U16	0 to 6553.5	0.1 V	

A22	Current maximum allowable discharging power	0x2F2E	RO	U16	0 to 65535	1 kW	
A23	Current maximum allowable charging power	0x2F2F	RO	U16	0 to 65535	1 kW	
A24	Reserved						
A25	Reserved						
A26	PCS integrated operation status	0x2F7D	RO	U16	0 to 65535	1	0: Stop 1: Standby 2: Grid-connected charging operation 3: Grid-connected discharging operation 4: Grid-connected zero power operation 5: Off-grid operation 6: Fault status
A27	PCS integrated operation status–bit mode	0x2F5A	RO	U16	0 to 65535		Bit 0: Stop Bit 1: Standby Bit 2: Grid-connected charging operation Bit 3: Grid-connected discharging operation Bit 4: Grid-connected zero power operation Bit 5: Off-grid operation Bit 6: Fault status Bit 7: Alarm status
A28	PCS fault status	0x2F43	RO	U16	0 to 1	1	0: Normal 1: Fault
A29	PCS alarm status	0x2F45	RO	U16	0 to 1	1	0: Normal 1: Alarm
A30	Charging operation status	0x2F3D	RO	U16	0 to 1	1	0: Non-charging operation 1: Charging operation
A31	Discharging operation status	0x2F3E	RO	U16	0 to 1	1	0: Non-discharging operation 1: Discharging operation
A32	Assembly operation status	0x2F40	RO	U16	0 to 1	1	0: Non-operating 1: Operating
A33	Battery voltage	0x2F48	RO	S16	–3276.8 to 3276.7	0.1 V	*The value is negative when the battery is connected reversely.*
A34	DC current	0x2F49	RO	S16	–32768 to 32767	1 A	-
A35	Grid frequency	0x2F4E	RO	U16	0 to 655.35	0.01 Hz	*Grid frequency obtained through phase-locked loop*
A36	DC power	0x2F4D	RO	S16	–32768 to 32767	1 kW	-
A37	Output power factor	0x2F53	RO	S16	–1.000 to 1.000	0.001	-
A38	Upper half bus voltage	0x2F46	RO	S16	–3276.8 to 3276.7	0.1 V	-
A39	Lower half bus voltage	0x2F47	RO	S16	–3276.8 to 3276.7	0.1 V	-

A40	Active power on the grid side	0x2F7E	RO	S16	–32768 to 32767	1 kW	-
A41	Reactive power on the grid side	0x2F7F	RO	S16	–32768 to 32767	1 kVar	-
A42	Total output apparent power	0x2F52	RO	S16	–32768 to 32767	1 kVA	-
A43	Phase AB grid line voltage	0x2F4F	RO	U16	0 to 6553.5	0.1 V	*Valid voltage value of grid line*
A44	Phase BC grid line voltage	0x2F50	RO	U16	0 to 6553.5	0.1 V	*Valid voltage value of grid line*
A45	Phase CA grid line voltage	0x2F51	RO	U16	0 to 6553.5	0.1 V	*Valid voltage value of grid line*
A46	Phase A current	0x2F4A	RO	U16	0 to 65535	1 A	*Valid current value*
A47	Phase B current	0x2F4B	RO	U16	0 to 65535	1 A	*Valid current value*
A48	Phase C current	0x2F4C	RO	U16	0 to 65535	1 A	*Valid current value*
A49	RMS value of inverter AB line voltage	0x2018	RO	U16	0 to 6553.5	0.1 V	*Phase AB output line voltage*
A50	RMS value of inverter BC line voltage	0x2019	RO	U16	0 to 6553.5	0.1 V	*Phase BC output line voltage*
A51	RMS value of inverter CA line voltage	0x201A	RO	U16	0 to 6553.5	0.1 V	*Phase CA output line voltage*
A52	Derated operation	0x2F59	RO	U16	0 to 1	1	Bit 0: Ambient temperature derating Bit 1: Coolant inlet temperature derating Bit 2: Fan derating Bit 3: Reactor temperature derating Bit 4: Master and slave derating together
A53	AC circuit breaker status	0x2F5B	RO	U16	0 to 1	1	0: Closed 1: Open
A54	AC pre-charge contactor status	0x2F5D	RO	U16	0 to 1	1	0: Closed 1: Open
A55	DC circuit breaker status (module 1)	0x2F5C	RO	U16	0 to 1	1	0: Closed 1: Open
A56	DC pre-charge contactor status (module 1)	0x2F5E	RO	U16	0 to 1	1	0: Closed 1: Open
A57	DC circuit breaker status (module 2)	0x2F84	RO	U16	0 to 1	1	0: Closed 1: Open
A58	DC pre-charge contactor status (module 2)	0x2F85	RO	U16	0 to 1	1	0: Closed 1: Open
A59	Emergency stop button status	0x2F60	RO	U16	0 to 65535	1	Bit 0: Local emergency stop signal Bit 1: Remote emergency stop signal Bit 2: BMS emergency stop signal

A60	A-phase IGBT temperature (module 1)	0x2F54	RO	S16	–3276.8 to 3276.7	0.1°C	-
A61	B-phase IGBT temperature (module 1)	0x2F55	RO	S16	–3276.8 to 3276.7	0.1°C	-
A62	C-phase IGBT temperature (module 1)	0x2F56	RO	S16	–3276.8 to 3276.7	0.1°C	-
A63	A-phase IGBT temperature (module 2)	0x2F80	RO	S16	–3276.8 to 3276.7	0.1°C	-
A64	B-phase IGBT temperature (module 2)	0x2F81	RO	S16	–3276.8 to 3276.7	0.1°C	-
A65	C-phase IGBT temperature (module 2)	0x2F82	RO	S16	–3276.8 to 3276.7	0.1°C	-
A66	Reactor temperature (module 1)	0x2F58	RO	S16	–3276.8 to 3276.7	0.1°C	-
A67	Reactor temperature (module 2)	0x2F83	RO	S16	–3276.8 to 3276.7	0.1°C	-
A68	Filter capacitor temperature	0x2028	RO	S16	–3276.8 to 3276.7	0.1°C	-
A69	Internal cavity temperature	0x2027	RO	S16	–3276.8 to 3276.7	0.1°C	-
A70	Ambient humidity	0x202F	RO	S16	–3276.8 to 3276.7	0.1% RH	-
A71	Ambient temperature	0x2F57	RO	S16	–3276.8 to 3276.7	0.1°C	-
A72	Fuse status (module 1)	0x2F5F	RO	U16	0 to 65535	1	Bit 0: DC fuse status (0: ON; 1: OFF) Bit 1: AC fuse status
A73	Fuse status (module 2)	0x2F86	RO	U16	0 to 65535	1	Bit 0: DC fuse status (0: ON; 1: OFF) Bit 1: AC fuse status
A74	Grounding insulation impedance (module 1)	0x2F6A	RO	U16	0 to 65535	1 kΩ	-
A75	Grounding insulation impedance (module 2)	0x2F87	RO	U16	0 to 65535	1 kΩ	-
A76	Water pump status	0x2F61	RO	U16	0 to 65535	1	0: Operating 1: Idle 2: Speed regulation signal absent 3: Locked-rotor/Overcurrent/Stopped *The first eight bits indicate the master water pump status. The last eight bits indicate the slave water pump status.*

A77	Three-phase fan status	0x2F63	RO	U16	0 to 65535	1	0: Normal 1: Fault
A78	DC fan status	0x2F62	RO	U16	0 to 65535	1	(0: Running; 1: Stopped) Bit 0: 48 V internal cavity fan 1 Bit 1: 48 V internal cavity fan 2 Bit 2: 48 V internal cavity fan 3 Bit 3: 48 V internal cavity fan 4 Bit 4: 48 V reactor fan 1 Bit 5: 48 V reactor fan 2 Bit 6: 24 V IGBT fan 1 Bit 7: 24 V IGBT fan 2 *The first eight bits indicate the master water pump status. The last eight bits indicate the slave water pump status.*
A79	DI status (module 1)	0x205B	RO	U16	0 to 65535	1	Bit 0: Door status (0: Closed; 1: Open) Bit 1: Insulation detection contactor status (0: Closed; 1: Open) Bit 2: Water level detection status (0: Normal; 1: Fault) Bit 3: DC surge protection device status (0: Normal; 1: Fault) Bit 4: AC surge protection device status (0: Normal; 1: Fault) Bit 5: Water pump power contactor status (0: Closed; 1: Open) Bit 6: Fan power contactor status (0: Closed; 1: Open) Bit 7: Heater contactor status (0: Closed; 1: Open) Bit 8: Reactor temperature control switch status (0: Normal temperature; 1: Over-temperature) Bit 9: DI9 standby Bit 10: DI11 standby Bit 11: DI12 standby Bit 12: DI13 standby Bit 13: DI14 standby Bit 14: DI15 standby Bit 15: DI16 standby
A80	DI status (module 2)	0x30C0	RO	U16	0 to 65535	1	Same as above
A81	Low accumulative discharge amount	0x2F64	RO	U16	0 to 65535	1 kWh	-
A82	High accumulative discharge amount	0x2F65	RO	U16	0 to 65535	1 kWh	-

A83	Low accumulative charge amount	0x2F66	RO	U16	0 to 65535	1 kWh	-
A84	High accumulative charge amount	0x2F67	RO	U16	0 to 65535	1 kWh	-
A85	Daily discharge amount	0x2F68	RO	U16	0 to 65535	1 kWh	-
A86	Daily charge amount	0x2F69	RO	U16	0 to 65535	1 kWh	-
A87	HMI version F	0x090F	RO	U16	0 to 655.35	0.01	
A88	HMI version F	0x0910	RO	U16	0 to 655.35	0.01	
A89	Customer code	0x0103	RO	U16	0 to 655.35	0.01	

4.2 BMS Parameters

Param.	Name	Address	Attribute	Data Type	Value Range	Unit	Description
B01	Running status of battery pack	0x0300	RO	U16	0 to 65535	1	Defined by the battery manufacturer
B02	Parameter_BMS status word	0x0301	RO	U16	0 to 65535	1	See Appendix 1 BMS Operation Status Word.
B03	Total voltage of battery pack	0x0302	RO	U16	0 to 6553.5	0.1 V	-
B04	Total current of battery pack	0x0303	RO	S16	–3276.8 to 3276.7	0.1 A	-
B05	Battery pack SOC	0x0304	RO	U16	0 to 100.0	0.1%	-
B06	Battery pack SOH	0x0305	RO	U16	0 to 100.0	0.1%	-
B07	Maximum charging current of battery pack	0x0306	RO	U16	0 to 6553.5	0.1 A	-
B08	Maximum discharging current of battery pack	0x0307	RO	U16	0 to 6553.5	0.1 A	-
B09	Maximum charging power of battery pack	0x0308	RO	U16	0 to 6553.5	0.1 kW	-
B10	Maximum discharging power of battery pack	0x0309	RO	U16	0 to 6553.5	0.1 kW	-
B11	Maximum SOC of battery cell	0x030A	RO	U16	0 to 100.0	0.1%	-
B12	Minimum SOC of battery cell	0x030B	RO	U16	0 to 100.0	0.1%	-
B13	Highest temperature of battery cell	0x030C	RO	S16	–100.0 to 3276.7	0.1°C	
B14	Lowest temperature of battery cell	0x030D	RO	S16	–100.0 to 3276.7	0.1°C	-
B15	Maximum voltage of battery cell	0x030E	RO	U16	0 to 65.535	0.001 V	
B16	Minimum voltage of battery cell	0x030F	RO	U16	0 to 65.535	0.001 V	-
B17	Upper voltage limit of battery pack	0x0310	RO	U16	0 to 6553.5	0.1 V	
B18	Lower voltage limit of	0x0311	RO	U16	0 to 6553.5	0.1 V	-

	battery pack						
B19	Available charge energy of battery pack	0x0312	RO	U16	0 to 6553.5	0.1 kWh	
B20	Available discharge energy of battery pack	0x0313	RO	U16	0 to 6553.5	0.1 kWh	-
B21	Rated energy of battery pack	0x0314	RO	U16	0 to 65.535	0.001 MWh	-
B22	Battery pack SOP	0x0315	RO	S16	0 to 6553.5	0.1 kW	-
B23	DC-side system status	0x0316	RO	S16	0 to 65535	1	-

4.3 Communication Parameters

Param.	Name	Address	Attribute	Data Type	Value Range	Unit	Description
C01	EMS communication protocol	0x0A00	RO	U16	0 to 4	1	0: Modbus RTU 1: Modbus TCP 2: IEC 61850-1 3: IEC 61850-2 4: EtherCAT
C02	EMS communication control	0x0A01	RO	U16	0 to 1	1	0: Disable 1: Enable
C03	EMS_Modbus local address	0x0A02	RO	U16	1 to 99	1	
C04	Reserved						
C05	Reserved						
C06	Reserved						
C07	Reserved						
C08	EMS_USART communication timeout interval	0x0A08	RO	U16	0 to 60	1s	The value 0 indicates that communication timeout is not detected.
C09	Number of EMS CRC error frames	0x0A09	RO	U16	0 to 65535	1	CRC check error *Modbus RTU and Ethernet expansion cards or IEC 61850 expansion card 1*
C10	EMS_UART_Number of overrun	0x0A0A	RO	U16	0 to 65535	1	-
C11	Expansion card communication timeout status	0x0A0B	RO	U16	0 to 65535	1	Bit 0: Card 1 external communication timeout Bit 1: Card 1 internal communication timeout (Inobus status) Bit 2: Card 2 external communication timeout Bit 3: Card 2 internal communication timeout (Inobus status) *Ethernet expansion card or IEC 61850 expansion card*
C12	Ethernet expansion card IP address 1	0x0A0C	RO	U16	0 to 255	1	
C13	Ethernet expansion card IP address 2	0x0A0D	RO	U16	0 to 255	1	
C14	Ethernet expansion card IP address 3	0x0A0E	RO	U16	0 to 255	1	

C15	Ethernet expansion card IP address 4	0x0A0F	RO	U16	2 to 255	1	
C16	Ethernet expansion card subnet mask 1	0x0A10	RO	U16	0 to 255	1	
C17	Ethernet expansion card subnet mask 2	0x0A11	RO	U16	0 to 255	1	
C18	Ethernet expansion card subnet mask 3	0x0A12	RO	U16	0 to 255	1	
C19	Ethernet expansion card subnet mask 4	0x0A13	RO	U16	0 to 255	1	
C20	Ethernet expansion card gateway address 1	0x0A14	RO	U16	0 to 255	1	
C21	Ethernet expansion card gateway address 2	0x0A15	RO	U16	0 to 255	1	
C22	Ethernet expansion card gateway address 3	0x0A16	RO	U16	0 to 255	1	
C23	Ethernet expansion card gateway address 4	0x0A17	RO	U16	0 to 255	1	
C24	Communication timeout interval of Ethernet expansion card	0x0A19	RO	U16	0 to 60	1s	The value 0 indicates that communication timeout is not detected. *Ethernet expansion card or IEC 61850 expansion card*
C25	Version of Ethernet expansion card 1	0x0A1A	RO	U16	0 to 655.35	0.01	*Ethernet expansion card or IEC 61850 expansion card 1*
C26	Ethernet communication error code	0x0A1B	RO	U16	0 to 65535	1	0: No error 1: External bus timeout *Ethernet expansion card or IEC 61850 expansion card 1*
C27	IEC 61850 expansion card 1_IP address 1	0x0A1C	RO	U16	0 to 255	1	
C28	IEC 61850 expansion card 1_IP address 2	0x0A1D	RO	U16	0 to 255	1	
C29	IEC 61850 expansion card 1_IP address 3	0x0A1E	RO	U16	0 to 255	1	
C30	IEC 61850 expansion card 1_IP address 4	0x0A1F	RO	U16	0 to 255	1	
C31	IEC 61850 expansion card 2_Version	0x0A20	RO	U16	0 to 655.35	0.01	
C32	IEC 61850 expansion card 2_Communication error code	0x0A21	RO	U16	0 to 65535	1	0: No error 1: External bus timeout
C33	IEC 61850 expansion card 2_Number of CRC error frames	0x0A22	RO	U16	0 to 65535	1	*CRC check error*
C34	IEC 61850 expansion card 2_IP address 1	0x0A23	RO	U16	0 to 255	1	
C35	IEC 61850 expansion card 2_IP address 2	0x0A24	RO	U16	0 to 255	1	
C36	IEC 61850 expansion card 2_IP address 3	0x0A25	RO	U16	0 to 255	1	
C37	IEC 61850 expansion card 2_IP address 4	0x0A26	RO	U16	0 to 255	1	

C38	EtherCAT communication error code	0x0A2B	RO	U16	0 to 65535	1	
C39	BMS communication protocol	0x0B00	RO	U16	0 to 2	1	0: Modbus RTU 1: Modbus TCP 2: CAN
C40	BMS communication control	0x0B01	RO	U16	0 to 1	1	0: Disable 1: Enable
C41	BMS Modbus local address	0x0B02	RO	U16	1 to 146	1	-
C42	BMS Modbus rate	0x0B03	RO	U16	0 to 6	1	0: 2400 1: 4800 2: 9600 3: 19200 4: 38400 5: 57600 6: 115200
C43	BMS Modbus data bit	0x0B04	RO	U16	0 to 1	1	0: 8 bits 1: 9 bits
C44	BMS Modbus check bit	0x0B05	RO	U16	0 to 2	1	0: None 1: Even parity 2: Odd parity
C45	BMS Modbus stop bit	0x0B06	RO	U16	0 to 1	1	0: 1 bit 1: 2 bits
C46	BMS Modbus communication timeout	0x0B08	RO	U16	0 to 60	1s	The value 0 indicates that communication timeout is not detected.
C47	Number of BMS Modbus CRC error frames	0x0B09	RO	U16	0 to 65535	1	*CRC check error*
C48	BMS UART overrun times	0x0B0A	RO	U16	0 to 65535	1	-
C49	BMS IP address 1	0x0B0C	RO	U16	0 to 255	1	-
C50	BMS IP address 2	0x0B0D	RO	U16	0 to 255	1	-
C51	BMS IP address 3	0x0B0E	RO	U16	0 to 255	1	-
C52	BMS IP address 4	0x0B0F	RO	U16	2 to 255	1	-
C53	BMS subnet mask address 1	0x0B10	RO	U16	0 to 255	1	-
C54	BMS subnet mask address 2	0x0B11	RO	U16	0 to 255	1	-
C55	BMS subnet mask address 3	0x0B12	RO	U16	0 to 255	1	-
C56	BMS subnet mask address 4	0x0B13	RO	U16	0 to 255	1	-
C57	BMS gateway address 1	0x0B14	RO	U16	0 to 255	1	-
C58	BMS gateway address 2	0x0B15	RO	U16	0 to 255	1	-
C59	BMS gateway address 3	0x0B16	RO	U16	0 to 255	1	-
C60	BMS gateway address 4	0x0B17	RO	U16	0 to 255	1	-
C61	Communication timeout of BMS Ethernet extension card	0x0B19	RO	U16	0 to 60	1s	The value 0 indicates that communication timeout is not detected.

C62	Version of BMS Ethernet extension card	0x0B1A	RO	U16	0 to 655.35	0.01	-
C63	BMS_Ethernet communication error code	0x0B1B	RO	U16	0 to 65535	1	0: No error 1: External bus timeout
C64	BMS_Ethernet expansion card communication timeout status	0x0B1C	RO	U16	0 to 65535	1	Bit 0: Card 1 external communication timeout Bit 1: Card 1 internal communication timeout (Inobus status) Bit 2: Reserved Bit 3: Reserved
C65	CAN communication rate	0x0B1E	RO	U16	0 to 4	1	0: 100 kbps 1: 125 kbps 2: 250 kbps 3: 500 kbps 4: 1 Mbps
C66	Reserved						
C67	BMS NMT node status	0x0B20	RO	U16	0 to 5	1	0: Initializing 1: Application layer reset 2: Communication reset 3: Pre-operational state 4: Operational state 5: Stopped
C68	BMS communication timeout	0x0B21	RO	U16	0 to 120	1s	The value 0 indicates that communication timeout is not detected.
C69	BMS CAN module status	0x0B22	RO	U16	0 to 2	1	0: Normal 1: Passive bus OFF 2: Bus OFF
C70	BMS CAN sending error count	0x0B23	RO	U16	0 to 255	1	-
C71	BMS CAN receiving error count	0x0B24	RO	U16	0 to 255	1	-
C72	Last BMS CAN LEC transmission error	0x0B25	RO	U16	0 to 7	1	0: No error 1: Stuff error 2: Format error 3: Response error 4: Recessive bit error 5: Dominant bit error 6: CRC error 7: Software configuration
C73	BMS CAN communication status	0x0B26	RO	U16	0 to 1	1	0: Normal communication 1: Timeout and no response

4.4 Fault Parameters

Param.	Name	Address	Attribute	Data Type	Value Range	Unit	Description
D01	Fault status word 1	0x2FB5	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D02	Fault status word 2	0x2FB6	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D03	Fault status word 3	0x2FB7	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D04	Fault status word 4	0x2FB8	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.

D05	Fault status word 5	0x2FB9	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D06	Fault status word 6	0x2FBA	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D07	Fault status word 7	0x2FBB	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D08	Fault status word 8	0x2FBC	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D09	Fault status word 9	0x2FBD	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D10	Fault status word 10	0x2FBE	RO	U16	0 to 65535	1	For details, see Appendix 2 PCS Fault Status Word.
D11	Alarm status word 1	0x2FBF	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D12	Alarm status word 2	0x2FC0	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D13	Alarm status word 3	0x2FC1	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D14	Alarm status word 4	0x2FC2	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D15	Alarm status word 5	0x2FC3	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D16	Alarm status word 6	0x2FC4	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D17	Alarm status word 7	0x2FC5	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.
D18	Alarm status word 8	0x2FC6	RO	U16	0 to 65535	1	For details, see Appendix 3 PCS Alarm Status Word.

* Note: Fault statuses are continuously updated; for faults not listed, refer to HMI display or consult PCS manufacturer.

5 Setting Parameters (Remote Control and Remote Regulation)

Param.	Name	Address	Attribute	Data Type	Value Range	Default	Unit	Description
S01	Command source	0x0E00	R/W	U16	0 to 2	0	1	0: HMI 1: EMS 2: Background (reserved)
S02	PCS operation mode	0x0E01	R/W	U16	0 to 2	0	1	0: On-grid 1: Off-grid
S03	PCS on-grid control mode	0x0E02	R/W	U16	0 to 2	0	1	0: P-Q mode 1: Constant DC current mode 2: Constant DC voltage mode
S04	Reactive power mode setting	0x0E04	R/W	U16	0 to 2	0	1	0: Constant reactive power 1: Constant PF 2: Q-U reactive
S05	Preferred power mode setting	0x0E06	R/W	U16	0 to 2	0	1	0: Proportional derating 1: Active power preferred 2: Reactive power preferred

S06	Active power setting	0x0E19	R/W	S16	K01[1] to K02[1]	0	1 kW	-
S07	Reactive power setting	0x0E1A	R/W	S16	K03[2] to K04[2]	0	1 kVar	-
S08	Power factor setting	0x0E1B	R/W	S16	–1.000 to 1.000	0	0.001	-
S09	Voltage setting in constant voltage mode	0x0E1C	R/W	U16	K05[3] to 1500.0	1300.0	0.1 V	*Sets the bus voltage (actual value) for constant-voltage charge/discharge in P-Q mode.*
S10	Current setting in constant current mode	0x0E1D	R/W	S16	K06[4] to K07[4]	0	1 A	*Sets the bus current (actual value) for constant-current charge/discharge in P-Q mode.*
S11	Power change rate setting	0x0E1E	R/W	U16	10 to 30000	10000	1% Pn/s	-
S12	Reserved							
S13	Off-grid acceleration rate	0x0E20	R/W	U16	0.01 to 1.00	0.50	0.01 pu/s	-
S14	Off-grid line voltage RMS	0x0E23	R/W	U16	0.0 to 1000.0	759.0	0.1 V	
S15	Insulation detection command	0x0E07	R/W	U16	0 to 1	0	1	0: Disable 1: Enable
S16	Start command	0x0E14	R/W	U16	0 to 1	0	1	0: Invalid 1: Send a start command
S17	Stop command	0x0E15	R/W	U16	0 to 1	0	1	0: Invalid 1: Send a stop command
S18	Fault reset	0x0E16	R/W	U16	0 to 1	0	1	0: Invalid 1: Send a reset command
S19	Standby command	0x0E17	R/W	U16	0 to 1	0	1	0: Invalid 1: Send a standby command
S20	Last synchronization time (YYYY)	0x0E27	R/W	U16	2000 to 2099	2020	1	*Provided to the EMS for system time synchronization*
S21	Last synchronization time (MMDD)	0x0E28	R/W	U16	101 to 1231	101	1	*Expressed in decimal format*
S22	Last synchronization time (hhmm)	0x0E29	R/W	U16	0 to 2359	0	1	-
S23	Last synchronization time (s)	0x0E2A	R/W	U16	0 to 59	0	1	-
S24	Reserved							

S25	Reserved							
S26	Reserved							
S27	Active silo detection and control function	0x057E	R/W	U16	0 to 1	0	1	0: Disable 1: Enable
S28	Zero power standby mode	0x0539	R/W	U16	0 to 1	1	1	0: Disable 1: Enable
S29	Reserved							
S30	Reserved							
S31	Reserved							
S32	Grid-forming function enable	0x1200	R/W	U16	0 to 1	0	1	0: Disable 1: Enable
S33	Primary frequency control enable	0x1208	R/W	U16	0 to 1	0	1	0: Disable 1: Enable
S34	Primary frequency control dead zone	0x120A	R/W	U16	0.00 to 0.10	0.03	0.01 Hz	
S35	Primary frequency control coefficient	0x1209	R/W	U16	0 to 65535	50	1	
S36	Primary voltage control enable	0x120C	R/W	U16	0 to 1	0	1	0: Disable 1: Enable
S37	Primary voltage control dead zone	0x120E	R/W	U16	0.00 to 0.10	0.00	0.01 pu	
S38	Primary voltage control coefficient	0x120D	R/W	U16	0.00 to 100.0	5.0	0.1	
S39	Virtual inertia coefficient	0x1203	R/W	U16	0 to 65535	100	1	
S40	Virtual damping coefficient	0x1204	R/W	U16	0 to 65535	200	1	
S41	Virtual excitation coefficient	0x1205	R/W	U16	0 to 65535	1200	1	

[1] K01 indicates the PCS internal parameter "Maximum charging power", and K02 indicates the PCS internal parameter "Maximum discharging power".

[2] K03 indicates the PCS internal parameter "Reactive power lower limit", and K04 indicates the PCS internal parameter "Reactive power upper limit".

[3] K05 indicates the PCS internal parameter "Lower voltage limit in constant voltage mode".

[4] K06 indicates the PCS internal parameter "Maximum charging current", and K07 indicates the PCS internal parameter "Maximum discharging current".

Appendix 1 BMS Operation Status Word

Group	Bit	Name	Description
BMS status word	0	BMS no communication	0: No fault; 1: Fault occurred
	1	BMS fault	0: No fault; 1: Fault occurred
	2	SOC above the upper limit (reserved)	0: No fault; 1: Fault occurred
	3	SOC below the lower limit (reserved)	0: No fault; 1: Fault occurred
	4	BMS charge prohibited	0: No fault; 1: Fault occurred
	5	BMS discharge prohibited	0: No fault; 1: Fault occurred
	6	Initializing	0: No fault; 1: Fault occurred
	7	BMS alarm	0: No fault; 1: Fault occurred
	8	BMS charge and discharge prohibited	0: No fault; 1: Fault occurred
	9	Reserved	0: No fault; 1: Fault occurred
	10	Reserved	0: No fault; 1: Fault occurred
	11	Reserved	0: No fault; 1: Fault occurred
	12	Reserved	0: No fault; 1: Fault occurred
	13	Reserved	0: No fault; 1: Fault occurred
	14	Reserved	0: No fault; 1: Fault occurred
	15	Reserved	0: No fault; 1: Fault occurred

[1] **When the BMS status word value is 0** (that is, all bits are 0), it indicates that the **BMS system is normal**.

Appendix 2 PCS Fault Status Word

Group	Bit	Fault Name	Description
Fault status word 1	0	Storage battery polarity connected reversely	0: No fault; 1: Fault occurred
	1	Storage battery voltage too high	0: No fault; 1: Fault occurred
	2	Storage battery voltage too low	0: No fault; 1: Fault occurred
	3	Battery voltage too high	0: No fault; 1: Fault occurred
	4	Battery voltage too low	0: No fault; 1: Fault occurred
	5	BMS system fault	0: No fault; 1: Fault occurred
	6	BMS pre-charge fault	0: No fault; 1: Fault occurred
	7	BMS main contactor fault	0: No fault; 1: Fault occurred
	8	BMS insulation fault	0: No fault; 1: Fault occurred
	9	DC side half-bus hardware overvoltage	0: No fault; 1: Fault occurred
	10	DC side full-bus software overvoltage	0: No fault; 1: Fault occurred
	11	DC side full-bus software undervoltage	0: No fault; 1: Fault occurred
	12	DC side half-bus voltage variation too large	0: No fault; 1: Fault occurred
	13	AC side hardware overcurrent	0: No fault; 1: Fault occurred
	14	AC side software overcurrent	0: No fault; 1: Fault occurred
	15	DC side hardware overcurrent	0: No fault; 1: Fault occurred
Fault status word 2	0	DC side software overcurrent	0: No fault; 1: Fault occurred
	1	IGBT Desat fault	0: No fault; 1: Fault occurred
	2	IGBT block fault	0: No fault; 1: Fault occurred
	3	IGBT over-temperature	0: No fault; 1: Fault occurred
	4	AC side A-phase Hall disconnected	0: No fault; 1: Fault occurred
	5	AC side B-phase Hall disconnected	0: No fault; 1: Fault occurred
	6	AC side C-phase Hall disconnected	0: No fault; 1: Fault occurred
	7	DC side current Hall disconnected	0: No fault; 1: Fault occurred
	8	Detection board model mismatch	0: No fault; 1: Fault occurred
	9	DC side pre-charge contactor fault	0: No fault; 1: Fault occurred
	10	DC side switch fault	0: No fault; 1: Fault occurred
	11	AC side pre-charge contactor fault	0: No fault; 1: Fault occurred
	12	AC side switch fault	0: No fault; 1: Fault occurred
	13	CBC overcurrent	0: No fault; 1: Fault occurred
	14	Device overload	0: No fault; 1: Fault occurred

	15	Grid AC overvoltage	0: No fault; 1: Fault occurred
Fault status word 3	0	Grid AC undervoltage	0: No fault; 1: Fault occurred
	1	Grid AC overfrequency	0: No fault; 1: Fault occurred
	2	Grid AC underfrequency	0: No fault; 1: Fault occurred
	3	Output phase loss	0: No fault; 1: Fault occurred
	4	Grid voltage sequence incorrect	0: No fault; 1: Fault occurred
	5	Phase lock error	0: No fault; 1: Fault occurred
	6	Grid voltage imbalance	0: No fault; 1: Fault occurred
	7	AC overvoltage in off-grid mode	0: No fault; 1: Fault occurred
	8	AC undervoltage in off-grid mode	0: No fault; 1: Fault occurred
	9	Islanding fault	0: No fault; 1: Fault occurred
	10	Current control deviation too large	0: No fault; 1: Fault occurred
	11	Bus soft-start failure	0: No fault; 1: Fault occurred
	12	DC component in the current beyond limit on the AC side	0: No fault; 1: Fault occurred
	13	AC switch closing voltage mismatch	0: No fault; 1: Fault occurred
	14	AD zero drift too large	0: No fault; 1: Fault occurred
	15	Low insulation resistance	0: No fault; 1: Fault occurred
Fault status word 4	0	Insulation detection failed	0: No fault; 1: Fault occurred
	1	Internal cavity over-temperature	0: No fault; 1: Fault occurred
	2	Internal cavity over-temperature alarm	0: No fault; 1: Fault occurred
	3	Internal cavity temperature low	0: No fault; 1: Fault occurred
	4	Filter capacitor over-temperature	0: No fault; 1: Fault occurred
	5	Filter capacitor over-temperature alarm	0: No fault; 1: Fault occurred
	6	Filter capacitor temperature low	0: No fault; 1: Fault occurred
	7	Auxiliary transformer over-temperature	0: No fault; 1: Fault occurred
	8	Auxiliary transformer over-temperature alarm	0: No fault; 1: Fault occurred
	9	Auxiliary transformer temperature low	0: No fault; 1: Fault occurred
	10	High ambient temperature	0: No fault; 1: Fault occurred
	11	High ambient temperature alarm	0: No fault; 1: Fault occurred
	12	Low ambient temperature	0: No fault; 1: Fault occurred
	13	IGBT module A-phase over-temperature	0: No fault; 1: Fault occurred
	14	IGBT module B-phase over-temperature	0: No fault; 1: Fault occurred
	15	IGBT module C-phase over-temperature	0: No fault; 1: Fault occurred
Fault status word 5	0	IGBT temperature sampling error	0: No fault; 1: Fault occurred

	1	IGBT over-temperature alarm	0: No fault; 1: Fault occurred
	2	Reactor over-temperature	0: No fault; 1: Fault occurred
	3	Reactor over-temperature alarm	0: No fault; 1: Fault occurred
	4	Reactor temperature low	0: No fault; 1: Fault occurred
	5	Surge protection device fault	0: No fault; 1: Fault occurred
	6	Parallel communication fault	0: No fault; 1: Fault occurred
	7	Abnormal stop of the master	0: No fault; 1: Fault occurred
	8	Carrier synchronization failed	0: No fault; 1: Fault occurred
	9	Internal cavity fan fault	0: No fault; 1: Fault occurred
	10	Circulating fan fault	0: No fault; 1: Fault occurred
	11	Reactor fan fault	0: No fault; 1: Fault occurred
	12	IGBT fan fault	0: No fault; 1: Fault occurred
	13	Reserved	0: No fault; 1: Fault occurred
	14	CAN communication fault between control board and communication board	0: No fault; 1: Fault occurred
	15	HMI communication fault	0: No fault; 1: Fault occurred
Fault status word 6	0	BMS communication fault	0: No fault; 1: Fault occurred
	1	Reserved	0: No fault; 1: Fault occurred
	2	Reserved	0: No fault; 1: Fault occurred
	3	Reserved	0: No fault; 1: Fault occurred
	4	Reserved	0: No fault; 1: Fault occurred
	5	EMS communication fault (dual network)	0: No fault; 1: Fault occurred
	6	EEPROM fault	0: No fault; 1: Fault occurred
	7	EEPROM alarm	0: No fault; 1: Fault occurred
	8	Black box error	0: No fault; 1: Fault occurred
	9	FPGA data exchange error	0: No fault; 1: Fault occurred
	10	RTC error	0: No fault; 1: Fault occurred
	11	220 V control power disconnected	0: No fault; 1: Fault occurred
	12	Leakage current Hall self-test failed	0: No fault; 1: Fault occurred
	13	Residual abrupt leakage current overlimit	0: No fault; 1: Fault occurred
	14	Continuous residual leakage current overlimit	0: No fault; 1: Fault occurred
	15	Local emergency stop fault	0: No fault; 1: Fault occurred
Fault status word 7	0	Remote emergency stop fault	0: No fault; 1: Fault occurred
	1	BMS emergency stop fault	0: No fault; 1: Fault occurred
	2	Phase-to-PE fault	0: No fault; 1: Fault occurred

	3	Capacitor voltage resonance fault	0: No fault; 1: Fault occurred
	4	High humidity	0: No fault; 1: Fault occurred
	5	Humidity detection disconnected	0: No fault; 1: Fault occurred
	6	Temperature too high	0: No fault; 1: Fault occurred
	7	Temperature detection disconnected	0: No fault; 1: Fault occurred
	8	Fuse fault	0: No fault; 1: Fault occurred
	9	Cabinet door fault	0: No fault; 1: Fault occurred
	10	Water pump locked-rotor	0: No fault; 1: Fault occurred
	11	Water pump speed regulation signal absent	0: No fault; 1: Fault occurred
	12	Water pump idling	0: No fault; 1: Fault occurred
	13	Water cooling system hydraulic fault	0: No fault; 1: Fault occurred
	14	Fan power sequence incorrect	0: No fault; 1: Fault occurred
	15	Fan power disconnected	0: No fault; 1: Fault occurred
Fault status word 8	0	Three-phase fan fault	0: No fault; 1: Fault occurred
	1	Heater contactor fault	0: No fault; 1: Fault occurred
	2	Water pump power disconnected	0: No fault; 1: Fault occurred
	3	Software version inconsistent between master and slave	0: No fault; 1: Fault occurred
	4	Dynamic environment fault	0: No fault; 1: Fault occurred
	5	DC side midpoint sampling disconnected	0: No fault; 1: Fault occurred
	6	DC side voltage sampling disconnected	0: No fault; 1: Fault occurred
	7	AC and DC voltage mismatch	0: No fault; 1: Fault occurred
	8	Grid voltage mismatch	0: No fault; 1: Fault occurred
	9	Heat exchanger fan fault	0: No fault; 1: Fault occurred
	10	DC circuit breaker fan fault	0: No fault; 1: Fault occurred
	11	AC circuit breaker fan fault	0: No fault; 1: Fault occurred
	12	Master and slave identification fault	0: No fault; 1: Fault occurred
	13	Power inconsistent between master and slave	0: No fault; 1: Fault occurred
	14	IGBT temperature low	0: No fault; 1: Fault occurred
	15	Zero sequence loop current too large	0: No fault; 1: Fault occurred
Fault status word 9	0	FPGA side fault	0: No fault; 1: Fault occurred
	1	Fault of parallel devices in cabinet	0: No fault; 1: Fault occurred
	2	Fiber communication interrupted	0: No fault; 1: Fault occurred
	3	Fiber communication CRC fault	0: No fault; 1: Fault occurred
	4	Control mode inconsistent	0: No fault; 1: Fault occurred

	5	Capacitor harmonic overlimit	0: No fault; 1: Fault occurred
	6	Fan pump and water pump switch error	0: No fault; 1: Fault occurred
	7	PCS interaction fault	0: No fault; 1: Fault occurred
	8	Capacitor harmonic sampling disconnected	0: No fault; 1: Fault occurred
	9	Reserved	0: No fault; 1: Fault occurred
	10	Reserved	0: No fault; 1: Fault occurred
	11	Reserved	0: No fault; 1: Fault occurred
	12	Reserved	0: No fault; 1: Fault occurred
	13	Reserved	0: No fault; 1: Fault occurred
	14	Reserved	0: No fault; 1: Fault occurred
	15	Reserved	0: No fault; 1: Fault occurred
	14	Reserved	0: No fault; 1: Fault occurred
	15	Reserved	0: No fault; 1: Fault occurred
Fault status word 10	0	Reserved	0: No fault; 1: Fault occurred
	1	Reserved	0: No fault; 1: Fault occurred
	2	Reserved	0: No fault; 1: Fault occurred
	3	Reserved	0: No fault; 1: Fault occurred
	4	Reserved	0: No fault; 1: Fault occurred
	5	Reserved	0: No fault; 1: Fault occurred
	6	Reserved	0: No fault; 1: Fault occurred
	7	Reserved	0: No fault; 1: Fault occurred
	8	Reserved	0: No fault; 1: Fault occurred
	9	Reserved	0: No fault; 1: Fault occurred
	10	Reserved	0: No fault; 1: Fault occurred
	11	Reserved	0: No fault; 1: Fault occurred
	12	Reserved	0: No fault; 1: Fault occurred
	13	Reserved	0: No fault; 1: Fault occurred
	14	Reserved	0: No fault; 1: Fault occurred
	15	Reserved	0: No fault; 1: Fault occurred

Appendix 3 PCS Alarm Status Word

Group	Bit	Fault Name	Description
Alarm status word 1	0	Storage battery polarity connected reversely	0: No alarm; 1: Alarm occurred
	1	Storage battery voltage too high	0: No alarm; 1: Alarm occurred
	2	Storage battery voltage too low	0: No alarm; 1: Alarm occurred
	3	Battery voltage too high	0: No alarm; 1: Alarm occurred
	4	Battery voltage too low	0: No alarm; 1: Alarm occurred
	5	BMS system fault	0: No alarm; 1: Alarm occurred
	6	BMS pre-charge fault	0: No alarm; 1: Alarm occurred
	7	BMS main contactor fault	0: No alarm; 1: Alarm occurred
	8	BMS insulation fault	0: No alarm; 1: Alarm occurred
	9	DC side half-bus hardware overvoltage	0: No alarm; 1: Alarm occurred
	10	DC side full-bus software overvoltage	0: No alarm; 1: Alarm occurred
	11	DC side full-bus software undervoltage	0: No alarm; 1: Alarm occurred
	12	DC side half-bus voltage variation too large	0: No alarm; 1: Alarm occurred
	13	AC side hardware overcurrent	0: No alarm; 1: Alarm occurred
	14	AC side software overcurrent	0: No alarm; 1: Alarm occurred
	15	DC side hardware overcurrent	0: No alarm; 1: Alarm occurred
Alarm status word 2	0	DC side software overcurrent	0: No alarm; 1: Alarm occurred
	1	IGBT Desat fault	0: No alarm; 1: Alarm occurred
	2	IGBT block fault	0: No alarm; 1: Alarm occurred
	3	IGBT over-temperature	0: No alarm; 1: Alarm occurred
	4	AC side A-phase Hall disconnected	0: No alarm; 1: Alarm occurred
	5	AC side B-phase Hall disconnected	0: No alarm; 1: Alarm occurred
	6	AC side C-phase Hall disconnected	0: No alarm; 1: Alarm occurred
	7	DC side current Hall disconnected	0: No alarm; 1: Alarm occurred
	8	Detection board model mismatch	0: No alarm; 1: Alarm occurred
	9	DC side pre-charge contactor fault	0: No alarm; 1: Alarm occurred
	10	DC side switch fault	0: No alarm; 1: Alarm occurred
	11	AC side pre-charge contactor fault	0: No alarm; 1: Alarm occurred
	12	AC side switch fault	0: No alarm; 1: Alarm occurred
	13	CBC overcurrent	0: No alarm; 1: Alarm occurred
	14	Device overload	0: No alarm; 1: Alarm occurred

	15	Grid AC overvoltage	0: No alarm; 1: Alarm occurred
Alarm status word 3	0	Grid AC undervoltage	0: No alarm; 1: Alarm occurred
	1	Grid AC overfrequency	0: No alarm; 1: Alarm occurred
	2	Grid AC underfrequency	0: No alarm; 1: Alarm occurred
	3	Output phase loss	0: No alarm; 1: Alarm occurred
	4	Grid voltage sequence incorrect	0: No alarm; 1: Alarm occurred
	5	Phase lock error	0: No alarm; 1: Alarm occurred
	6	Grid voltage imbalance	0: No alarm; 1: Alarm occurred
	7	AC overvoltage in off-grid mode	0: No alarm; 1: Alarm occurred
	8	AC undervoltage in off-grid mode	0: No alarm; 1: Alarm occurred
	9	Islanding fault	0: No alarm; 1: Alarm occurred
	10	Current control deviation too large	0: No alarm; 1: Alarm occurred
	11	Bus soft-start failure	0: No alarm; 1: Alarm occurred
	12	DC component in the current beyond limit on the AC side	0: No alarm; 1: Alarm occurred
	13	AC switch closing voltage mismatch	0: No alarm; 1: Alarm occurred
	14	AD zero drift too large	0: No alarm; 1: Alarm occurred
	15	Low insulation resistance	0: No alarm; 1: Alarm occurred
Alarm status word 4	0	Insulation detection failed	0: No alarm; 1: Alarm occurred
	1	Internal cavity over-temperature	0: No alarm; 1: Alarm occurred
	2	Internal cavity over-temperature alarm	0: No alarm; 1: Alarm occurred
	3	Internal cavity temperature low	0: No alarm; 1: Alarm occurred
	4	Filter capacitor over-temperature	0: No alarm; 1: Alarm occurred
	5	Filter capacitor over-temperature alarm	0: No alarm; 1: Alarm occurred
	6	Filter capacitor temperature low	0: No alarm; 1: Alarm occurred
	7	Auxiliary transformer over-temperature	0: No alarm; 1: Alarm occurred
	8	Auxiliary transformer over-temperature alarm	0: No alarm; 1: Alarm occurred
	9	Auxiliary transformer temperature low	0: No alarm; 1: Alarm occurred
	10	High ambient temperature	0: No alarm; 1: Alarm occurred
	11	High ambient temperature alarm	0: No alarm; 1: Alarm occurred
	12	Low ambient temperature	0: No alarm; 1: Alarm occurred
	13	IGBT module A-phase over-temperature	0: No alarm; 1: Alarm occurred
	14	IGBT module B-phase over-temperature	0: No alarm; 1: Alarm occurred
	15	IGBT module C-phase over-temperature	0: No alarm; 1: Alarm occurred
Alarm status word 5	0	IGBT temperature sampling error	0: No alarm; 1: Alarm occurred

	1	IGBT over-temperature alarm	0: No alarm; 1: Alarm occurred
	2	Reactor over-temperature	0: No alarm; 1: Alarm occurred
	3	Reactor over-temperature alarm	0: No alarm; 1: Alarm occurred
	4	Reactor temperature low	0: No alarm; 1: Alarm occurred
	5	Surge protection device fault	0: No alarm; 1: Alarm occurred
	6	Parallel communication fault	0: No alarm; 1: Alarm occurred
	7	Abnormal stop of the master	0: No alarm; 1: Alarm occurred
	8	Carrier synchronization failed	0: No alarm; 1: Alarm occurred
	9	Internal cavity fan fault	0: No alarm; 1: Alarm occurred
	10	Circulating fan fault	0: No alarm; 1: Alarm occurred
	11	Reactor fan fault	0: No alarm; 1: Alarm occurred
	12	IGBT fan fault	0: No alarm; 1: Alarm occurred
	13	Reserved	0: No alarm; 1: Alarm occurred
	14	CAN communication fault between control board and communication board	0: No alarm; 1: Alarm occurred
	15	HMI communication fault	0: No alarm; 1: Alarm occurred
Alarm status word 6	0	BMS communication fault	0: No alarm; 1: Alarm occurred
	1	Reserved	0: No alarm; 1: Alarm occurred
	2	Reserved	0: No alarm; 1: Alarm occurred
	3	Reserved	0: No alarm; 1: Alarm occurred
	4	Reserved	0: No alarm; 1: Alarm occurred
	5	EMS communication fault (dual network)	0: No alarm; 1: Alarm occurred
	6	EEPROM fault	0: No alarm; 1: Alarm occurred
	7	EEPROM alarm	0: No alarm; 1: Alarm occurred
	8	Black box error	0: No alarm; 1: Alarm occurred
	9	FPGA data exchange error	0: No alarm; 1: Alarm occurred
	10	RTC error	0: No alarm; 1: Alarm occurred
	11	220 V control power disconnected	0: No alarm; 1: Alarm occurred
	12	Leakage current Hall self-test failed	0: No alarm; 1: Alarm occurred
	13	Residual abrupt leakage current overlimit	0: No alarm; 1: Alarm occurred
	14	Continuous residual leakage current overlimit	0: No alarm; 1: Alarm occurred
	15	Local emergency stop fault	0: No alarm; 1: Alarm occurred
Alarm status word 7	0	Remote emergency stop fault	0: No alarm; 1: Alarm occurred
	1	BMS emergency stop fault	0: No alarm; 1: Alarm occurred
	2	Phase-to-PE fault	0: No alarm; 1: Alarm occurred

	3	Capacitor voltage resonance fault	0: No alarm; 1: Alarm occurred
	4	High humidity	0: No alarm; 1: Alarm occurred
	5	Humidity detection disconnected	0: No alarm; 1: Alarm occurred
	6	Temperature too high	0: No alarm; 1: Alarm occurred
	7	Temperature detection disconnected	0: No alarm; 1: Alarm occurred
	8	Fuse fault	0: No alarm; 1: Alarm occurred
	9	Cabinet door fault	0: No alarm; 1: Alarm occurred
	10	Water pump locked-rotor	0: No alarm; 1: Alarm occurred
	11	Water pump speed regulation signal absent	0: No alarm; 1: Alarm occurred
	12	Water pump idling	0: No alarm; 1: Alarm occurred
	13	Water cooling system hydraulic fault	0: No alarm; 1: Alarm occurred
	14	Fan power sequence incorrect	0: No alarm; 1: Alarm occurred
	15	Fan power disconnected	0: No alarm; 1: Alarm occurred
Alarm status word 8	0	Three-phase fan fault	0: No alarm; 1: Alarm occurred
	1	Heater contactor fault	0: No alarm; 1: Alarm occurred
	2	Water pump power disconnected	0: No alarm; 1: Alarm occurred
	3	Software version inconsistent between master and slave	0: No alarm; 1: Alarm occurred
	4	Dynamic environment fault	0: No alarm; 1: Alarm occurred
	5	DC side midpoint sampling disconnected	0: No alarm; 1: Alarm occurred
	6	DC side voltage sampling disconnected	0: No alarm; 1: Alarm occurred
	7	AC and DC voltage mismatch	0: No alarm; 1: Alarm occurred
	8	Grid voltage mismatch	0: No alarm; 1: Alarm occurred
	9	Heat exchanger fan fault	0: No alarm; 1: Alarm occurred
	10	DC circuit breaker fan fault	0: No alarm; 1: Alarm occurred
	11	AC circuit breaker fan fault	0: No alarm; 1: Alarm occurred
	12	Master and slave identification fault	0: No alarm; 1: Alarm occurred
	13	Power inconsistent between master and slave	0: No alarm; 1: Alarm occurred
	14	IGBT temperature low	0: No alarm; 1: Alarm occurred
	15	Zero sequence loop current too large	0: No alarm; 1: Alarm occurred

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